

Higher-order association rules in gastrointestinal aGVHD

Proof of added information, atlas alignment, and new biological hypotheses

How to read this report. A higher-order rule contains more than two cell types. A rule is *new* in one FOV when it passes rule-level $FDR \leq 0.05$ but none of its constituent pair rules was mined in that FOV. New attractions require $Lift \geq 1.3$; new avoidances require inverse Lift, $1/Lift, \geq 1.3$. Stage analyses require at least 20 cells of every rule cell type and at least five rule-bearing FOVs in one stage. For a new rule, the pair comparison is a lower bound based on the pair-mining threshold, not a measured pair-to-complex ratio. Individual FOVs have rule-level statistical support; stage prevalence and patient-median Lift are descriptive and are not patient-level significance tests. Spatial association does not establish causality.

1 Proof that higher-order mining adds information

The first test is global: after restricting to rules with $FDR \leq 0.05$, does the method mostly repeat shorter rules, or does it isolate a visible, structured remainder? Most long rules are correctly labeled redundant, but the nonredundant fraction is substantial and depends on rule shape. For attractions, 5–21% are clearly stronger than a mined parent and 6–12% have no mined parent. For avoidances, clearly stronger rules reach 27% for one-to-many rules. This separation is the method proof: the algorithm does not claim that every longer rule is useful; it distinguishes inherited pair effects from joint contexts.

Evidence strength: Strong as an internal method check; these are rule counts, not biological effect sizes.

What the longer rules add

Statistically supported rules
attracts · $FDR \leq 0.05$

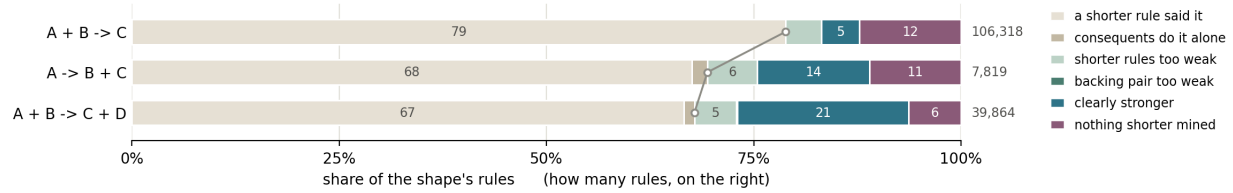


Figure 1: Verdicts among statistically supported attractive rules. Percentages are within rule shape; totals at right show that the result is not driven by a small rule family.

What the longer rules add

Statistically supported rules
avoids · $FDR \leq 0.05$

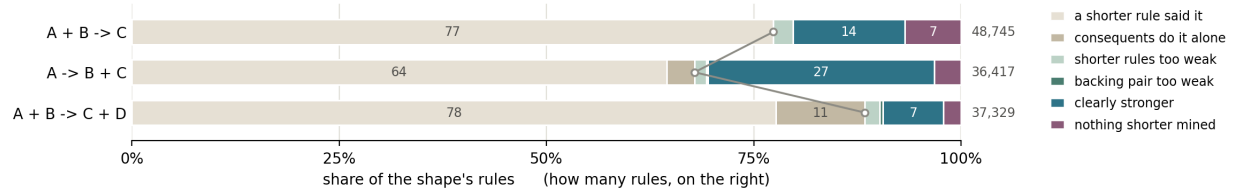


Figure 2: The same classification for statistically supported avoidance rules. The result confirms that both spatial directions contain nonredundant higher-order information, while most longer rules are still rejected as shorter-rule restatements.

1.1 Two direct ways a complex rule can beat its simpler rule

The strongest direct example is Endothelial, Epithelial $\rightarrow$ Goblet, SMV in a Colon FOV. Its Lift is 23.3 versus 4.45 for the strongest simpler rule, a measured 5.2 $\times$ gain. The second example is different: adding Epithelial to Endothelial, Macrophage $\rightarrow$ CD8T raises Lift only from 1.34 to 1.45, but changes the FDR from 0.743 to 0.00189. Thus higher-order mining can add either a much larger effect or a joint context in which an otherwise unsupported relationship becomes statistically detectable.

Evidence strength: Strong at the displayed FOV level; examples are selected automatically by the notebook.

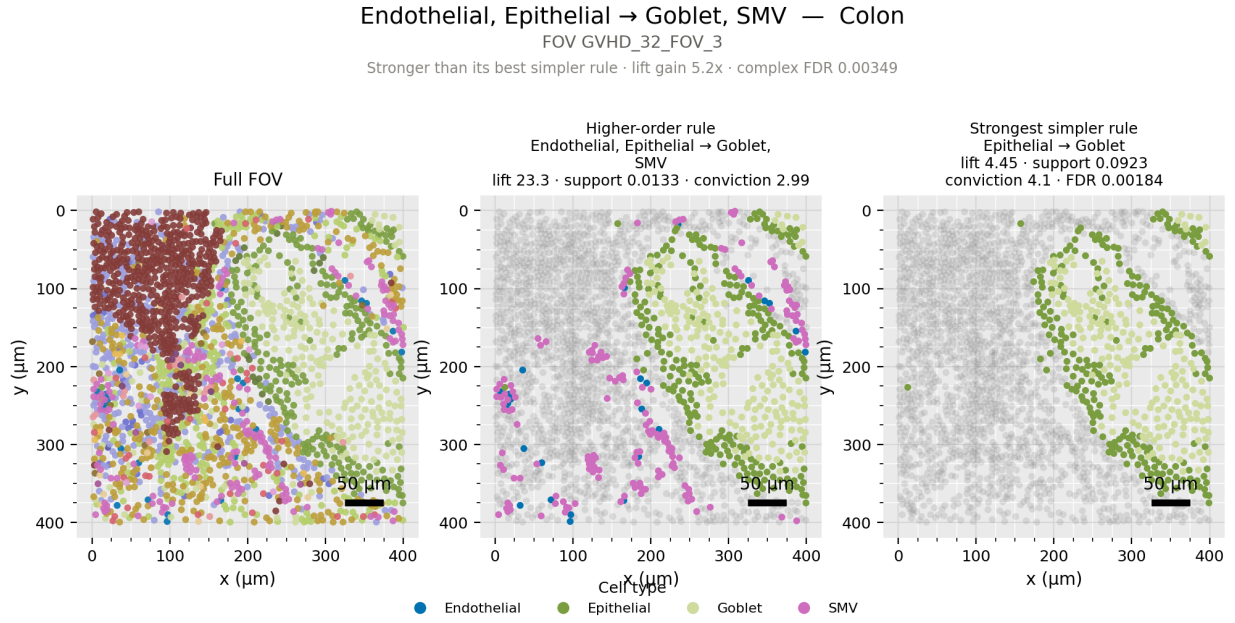


Figure 3: A measured stronger-effect example. The complete FOV, higher-order rule and strongest mined simpler rule use one color map and print Lift, support, conviction and FDR where available.

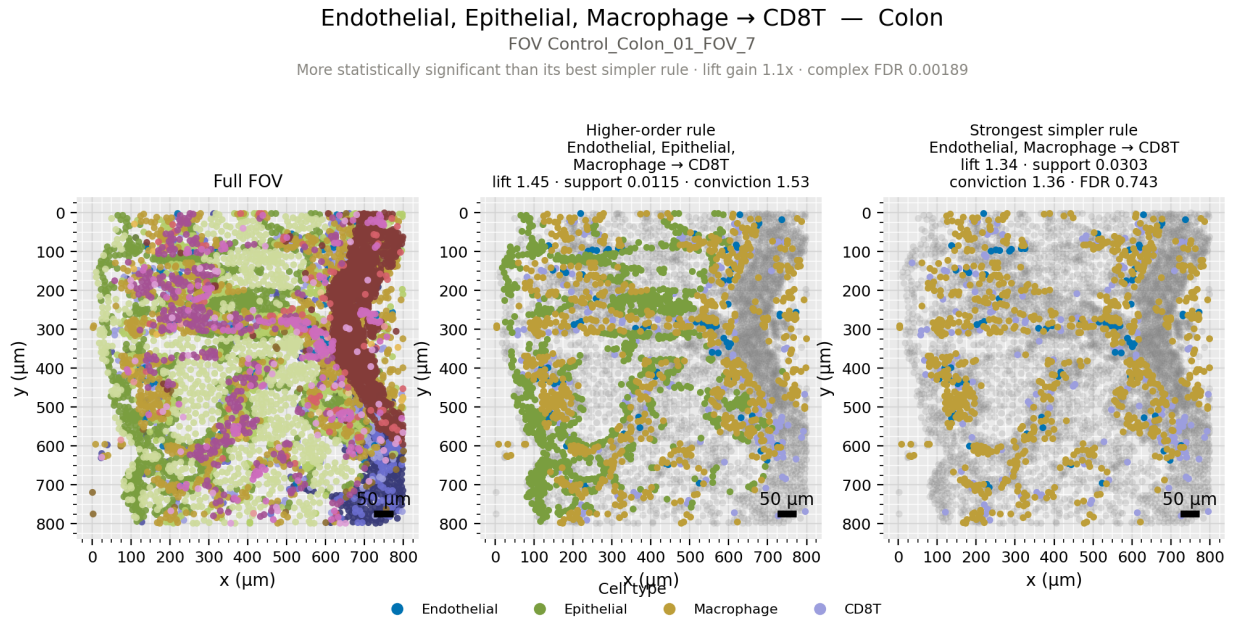


Figure 4: A measured significance-gain example. The additional Epithelial context converts the strongest parent from unsupported to rule-level FDR < 0.05 while preserving a modest Lift gain.

1.2 New rules recur when no pair rule is available

New rules provide the strictest proof of added representational capacity: no constituent pair was mined in the same FOV. The strongest recurring attractive families include epithelial–plasma, stromal–vascular and neural–epithelial contexts. Avoidances also recur, although their lower-bound gains are generally smaller.

New higher-order attraction	FOVs	median Lift	gain floor
Epithelial, Macrophage → Plasma	24	1.45	$\geq 1.21\times$
Epithelial, SMV → Plasma	22	1.43	$\geq 1.19\times$
Epithelial, Fibroblast → Plasma	18	1.43	$\geq 1.19\times$
Epithelial, SMV → Endothelial	13	2.76	$\geq 2.30\times$
Epithelial, Neuron → Paneth	12	2.24	$\geq 1.86\times$

Table 1: Representative recurrent new attractions, ordered primarily by rule-bearing FOVs. The gain floor compares the complex Lift with the largest value allowed for an unmined pair.

New higher-order avoidance	FOVs	median inverse Lift	gain floor
Muscle → Fibroblast, Macrophage	17	1.40	$\geq 1.12\times$
Fibroblast, Neuron → Goblet	13	1.45	$\geq 1.16\times$
Endothelial, Fibroblast → Goblet	12	1.66	$\geq 1.33\times$
Muscle, SMV → Fibroblast	12	1.43	$\geq 1.15\times$
Muscle, SMV → Macrophage	11	1.44	$\geq 1.15\times$

Table 2: Representative recurrent new avoidances. Inverse Lift puts attraction and avoidance strength on the same “larger is stronger” scale.

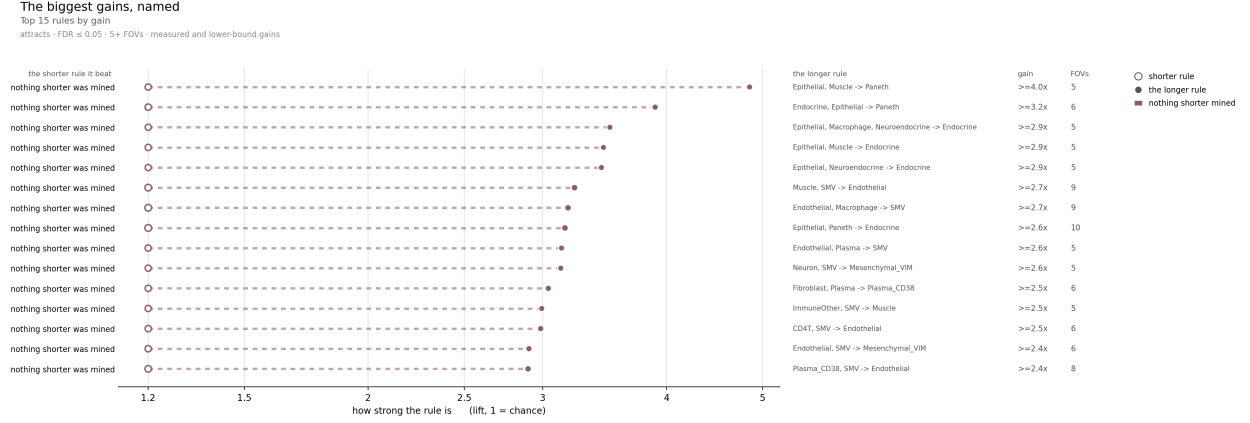


Figure 5: The largest lower-bound gains among recurring new attractions. Every rule passes FDR 0.05, Lift 1.3 and the five-FOV recurrence filter.

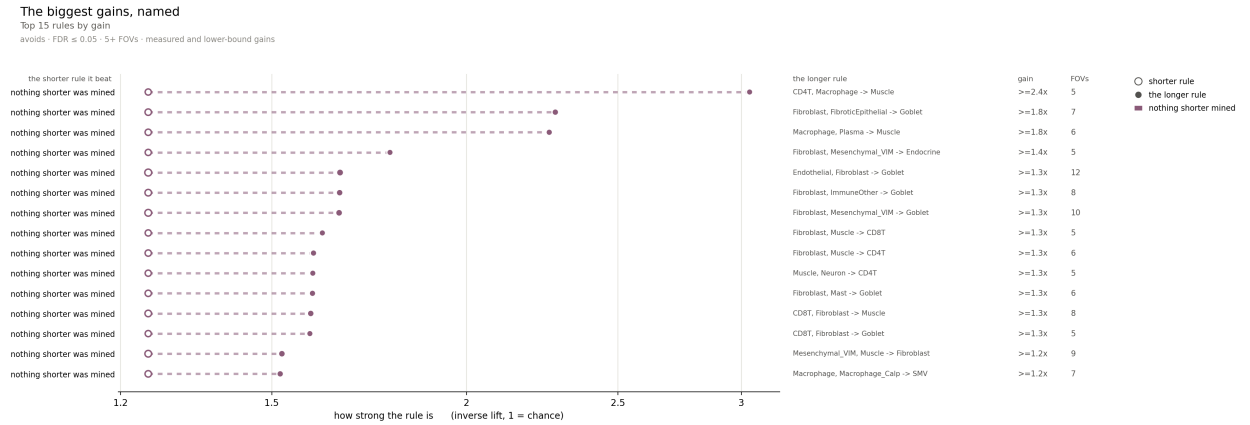


Figure 6: The corresponding ranking for new avoidances, using inverse Lift. The recurrent set is real but less strongly separated from the pair-mining boundary than the leading attractions.

1.3 The stage result is not explained only by cell availability

A non-new Colon rule provides a controlled comparison because its strongest simpler rule is measurable. Goblet, Macrophage → Epithelial, Plasma falls from 48% of Control FOVs to 2% of Mild and 9% of Severe FOVs when all FOVs form the denominator. After requiring 20 cells of every rule type, the complex prevalences are 48%, 8% and 21%, while the strongest simpler rule remains much more common at 92%, 58% and 79%. Eligibility changes the denominator but does not erase the loss of full joint organization.

Evidence strength: Descriptive availability control; not a patient-level stage test.

Figure 7 displays the missing control directly. Every stage FOV is partitioned into too few cells to test, eligible but empty, covered by a simpler rule, or carrying the informative complex rule. Thus the reader can see cell availability and spatial organization in the same denominator rather than infer one from two separate percentages.

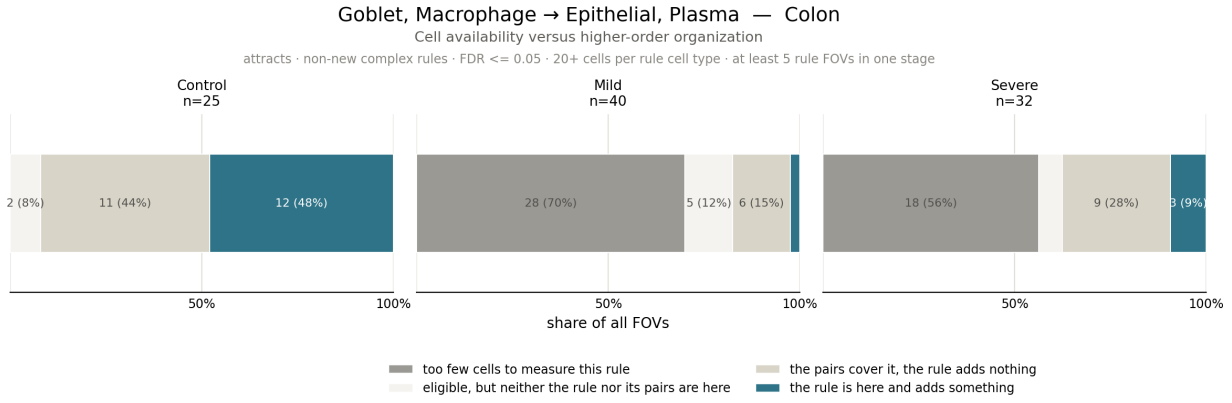


Figure 7: Every Colon FOV is counted exactly once. Grey is insufficient cell availability; the other segments are eligible FOVs split by what spatial rule was found. The informative-rule segment shrinks in disease even after the unavailable FOVs are made explicit.

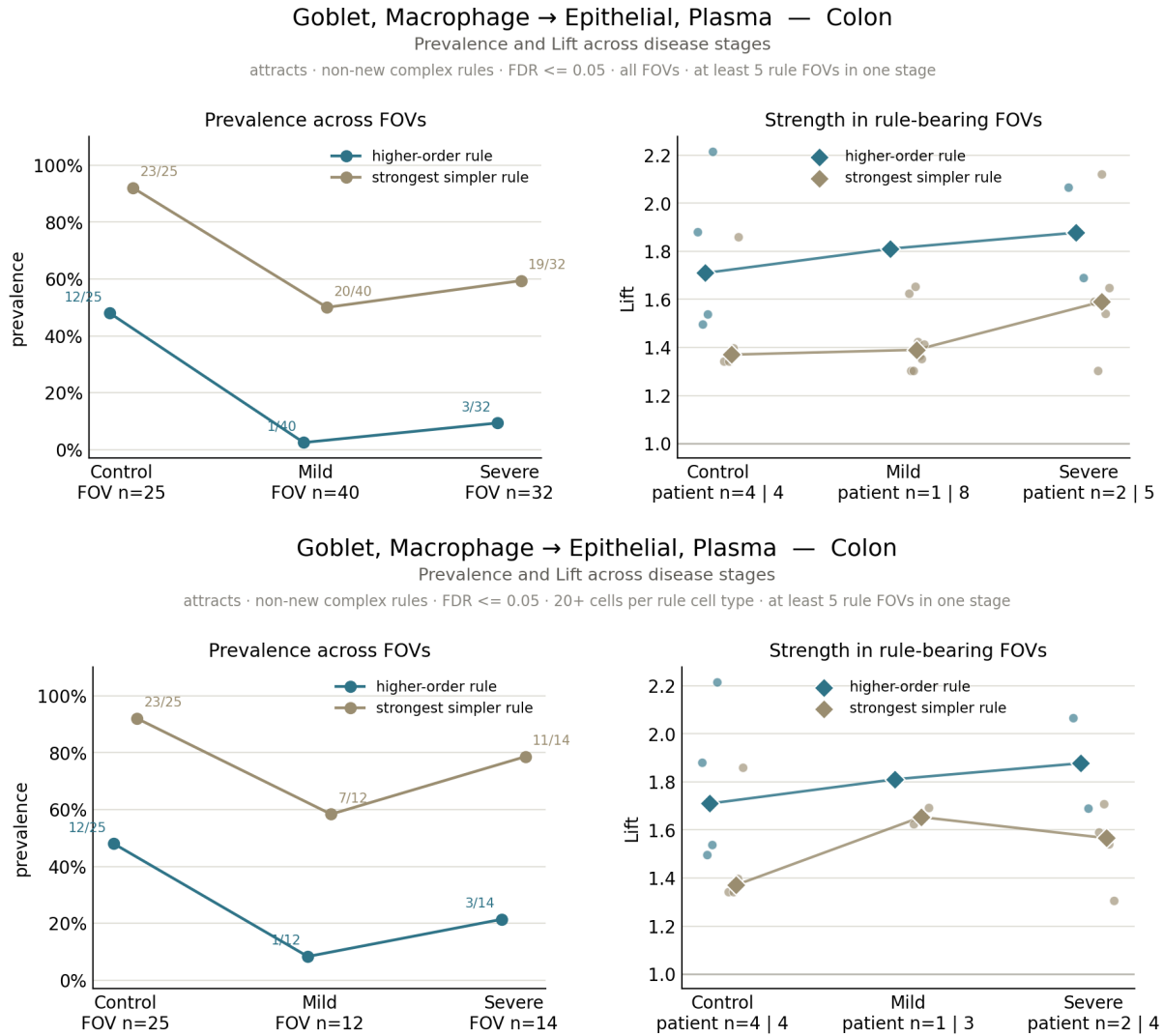


Figure 8: The same Colon rule before (upper) and after (lower) cell-eligibility filtering. Both prevalence lines compare the complex and strongest simpler rule; the right panels compare their patient-level Lift in rule-bearing FOVs.

The aggregate comparison establishes the stage pattern; Figure 9 then makes the selected tissue examples inspectable rather than asking the reader to trust the summary alone.

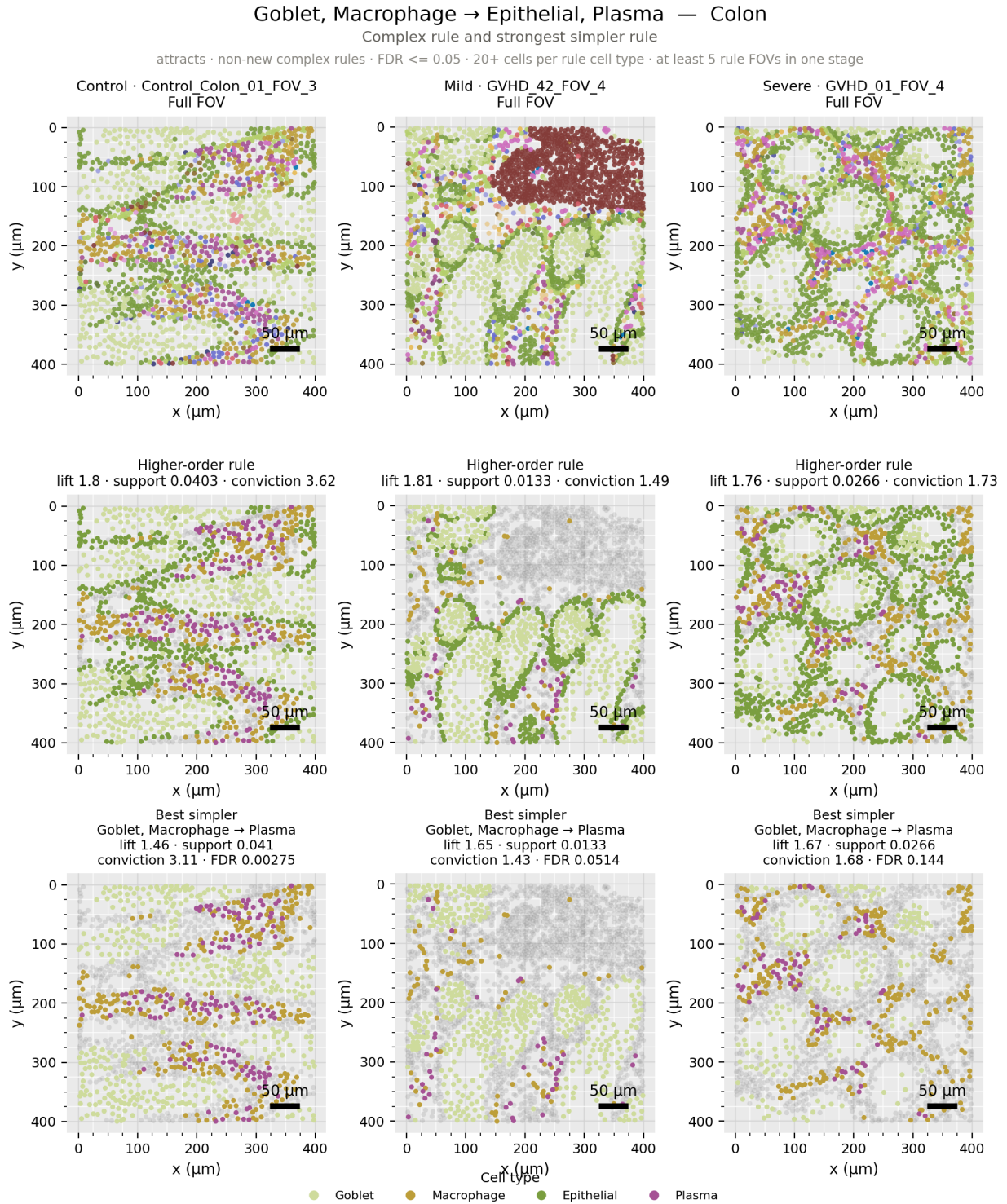


Figure 9: Eligible Control, Mild and Severe examples for the availability check. Each column shows the full FOV, the complex rule and its strongest simpler rule.

1.4 Complex and simpler rules can move in opposite directions

The method should also expose cases in which the longer organization changes differently from its parts. Figure 10 provides a filtered example: Epithelial, Neuron → Goblet, Macrophage becomes more prevalent from Control to Severe while the constituent simpler-rule coverage decreases slightly. In rule-bearing FOVs, the complex Lift decreases from 2.86 to 2.43 while the simpler-rule Lift increases from 1.91 to 2.21. The two panels therefore show genuine opposite movement in both summaries, not a duplicated Lift axis.

Evidence strength: Descriptive reversal passing the five-FOV recurrence filter; patient counts are printed in the figure.

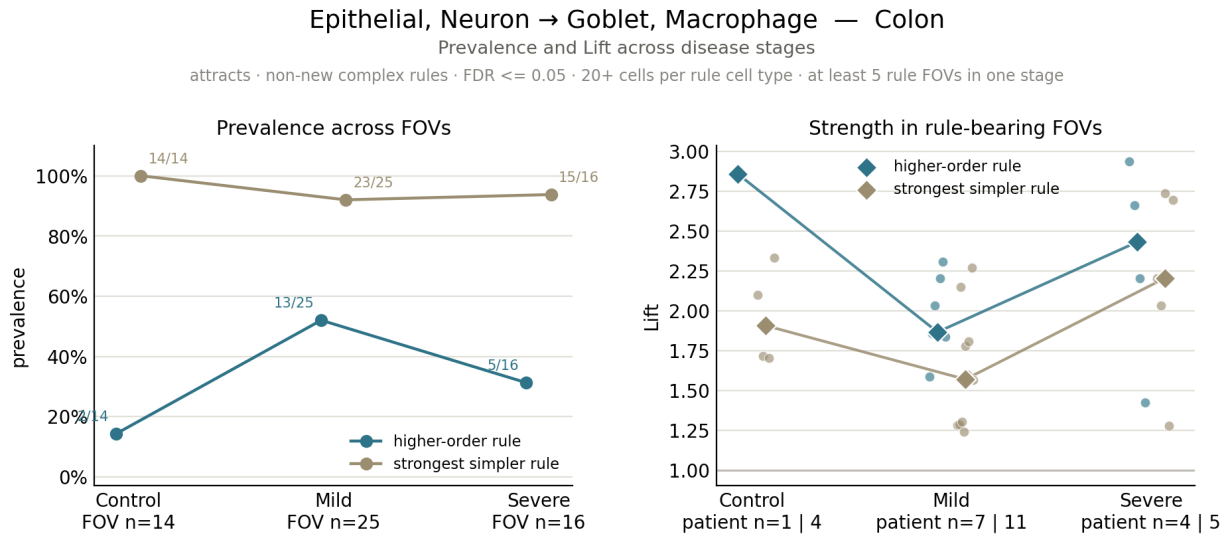


Figure 10: An explicit simple-versus-complex reversal. The left panel compares prevalence across the same eligible stage FOVs; the right compares patient-median Lift only where each rule is observed.

The same screen found no new-only rule with opposite complex-versus-simpler prevalence movement between any two stages after the five-FOV filter. This is expected to be harder: for a new occurrence, no constituent pair is mined in that same FOV, so the only possible comparison is with pair-rule prevalence elsewhere in the stage. No weaker example is shown.

2 Higher-order results aligned with the spatial-atlas paper

2.1 Loss of organized plasma-cell contexts

The [spatial atlas](#) reports that “CD4 T cells, plasma cells, and macrophages were spatially organized from the crypts to the tip of the villus” and that homeostatic immune organization is disrupted with a prominent reduction in IgA-secreting plasma cells. The complex-rule screen independently recovers a family of Colon plasma contexts. Figure 11 tests whether this is one selected rule or a coordinated family; Figure 12 then tests whether the leading rule disappears merely because its cell types become unavailable.

Evidence strength: Strong descriptive family-level alignment; stage contrasts are not permutation-tested here.

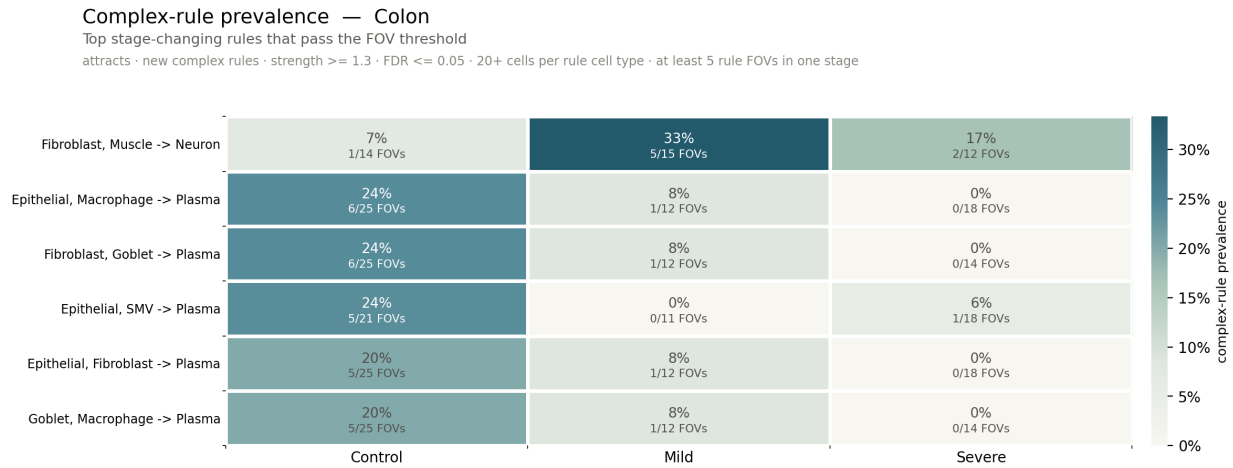


Figure 11: All recurring new Colon attractions that pass the current filters. Five of six rows end in Plasma and show a coordinated loss from Control. Exact eligible denominators are printed in every cell.

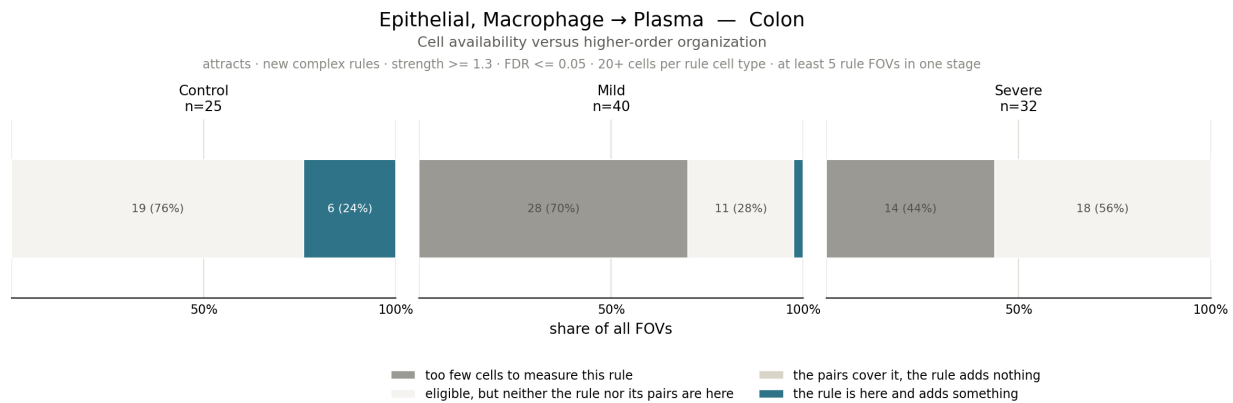


Figure 12: Cell availability and spatial organization for the leading Colon plasma rule. Each stage bar uses all FOVs: grey marks too few cells, while the plum segment marks eligible FOVs with the new complex rule. The loss of the plum segment is visible separately from changing cell availability.

Together, the figures show five related new rules ending in Plasma, with 20–24% prevalence in eligible Control FOVs but at most 8% in Mild and 6% in Severe. Epithelial, Macrophage → Plasma is also the most recurrent new rule overall (24 FOVs; median Lift 1.45). It refines rather than repeats the atlas result because no constituent pair was mined in those rule-bearing FOVs. Figure 13 shows the automatically selected Colon examples behind that summary.

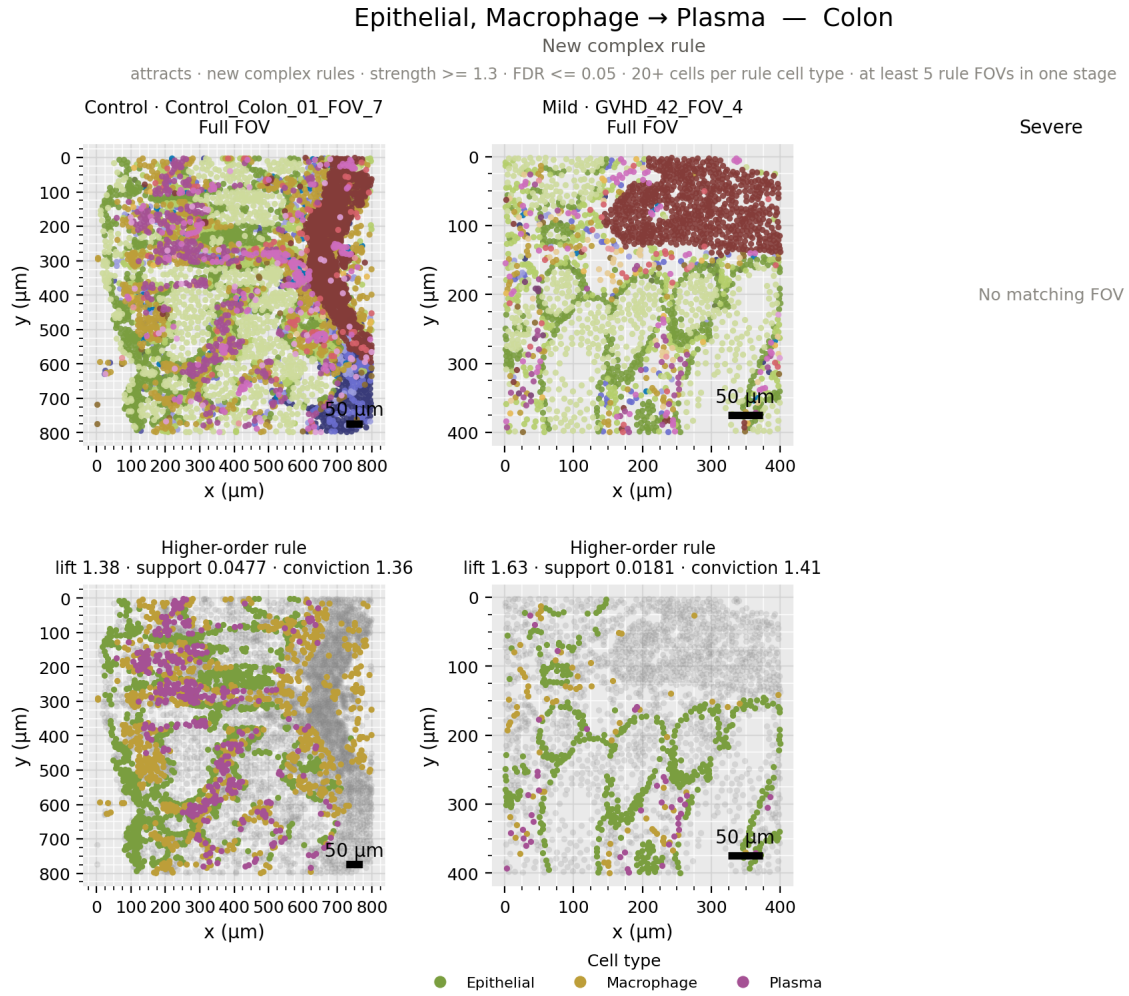


Figure 13: Stage-specific Colon FOV examples for Epithelial, Macrophage → Plasma. Because the rule is new, each available column shows the full FOV and the highlighted higher-order rule without inventing a simpler comparator.

2.2 Stromal–vascular remodeling is specifically higher-order

The atlas describes increased fibrosis and broad stromal disruption. In eligible Colon FOVs, the screen identifies Endothelial, Epithelial → Goblet, SMV as the clearest Severe-enriched non-new attraction. The next three figures expose the complete argument: its position in the filtered screen, the availability-versus-organization partition, and the direct complex-versus-simpler prevalence and Lift comparison.

Evidence strength: Strong descriptive separation from the simpler rule; Severe includes five complex-rule patients.

Complex-rule prevalence — Colon

Top stage-changing rules that pass the FOV threshold

attracts · non-new complex rules · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

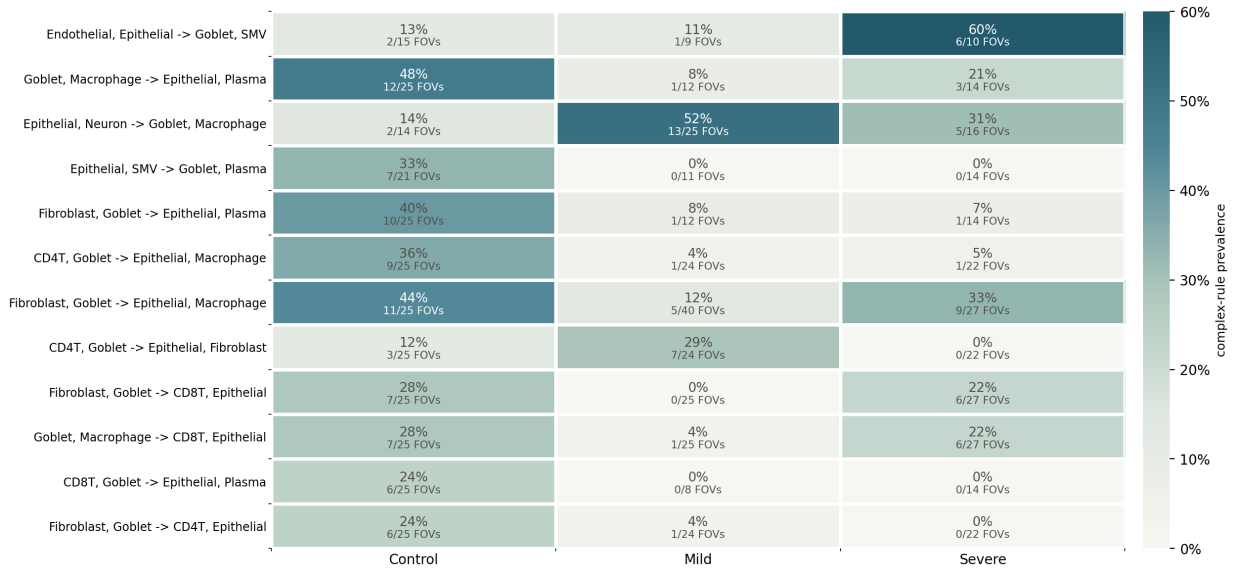


Figure 14: Top eligible non-new Colon attractions. The stromal-vascular rule is the clearest Severe-enriched pattern; the same overview also shows the Control-enriched plasma contexts.

Endothelial, Epithelial → Goblet, SMV — Colon

Cell availability versus higher-order organization

attracts · non-new complex rules · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

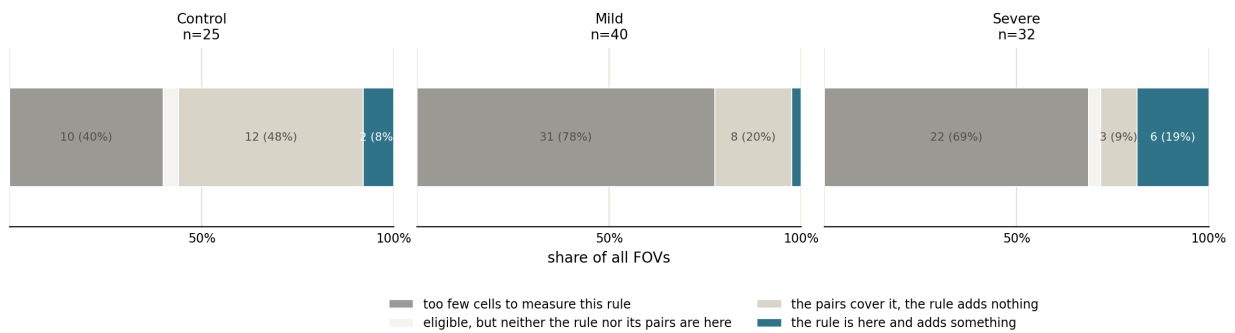


Figure 15: All Colon FOVs partitioned by whether the four required cell types are sufficiently available and, if so, whether the FOV contains neither rule, only simpler coverage, or the informative complex rule. The Severe enrichment is visible in the final segment and cannot be read as cell availability alone.

Endothelial, Epithelial → Goblet, SMV — Colon

Prevalence and Lift across disease stages

attracts · non-new complex rules · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

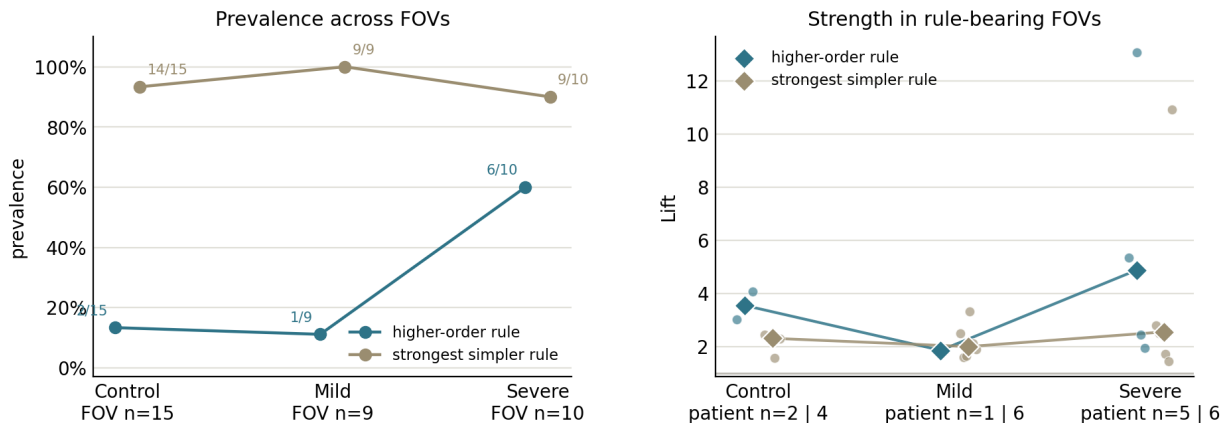


Figure 16: Prevalence and Lift for Endothelial, Epithelial → Goblet, SMV and its strongest simpler rule. The complex rule reaches 6/10 eligible Severe FOVs; patient counts are printed below the Lift panel.

Figures 15–16 show the numerical separation: complex prevalence rises from 13% and 11% to 60%, while simpler-rule coverage changes in the opposite direction from 93% to 90%. The complex Lift also exceeds the simpler Lift in the Control and Severe medians. The selected maps below show what the added four-cell context looks like in tissue.

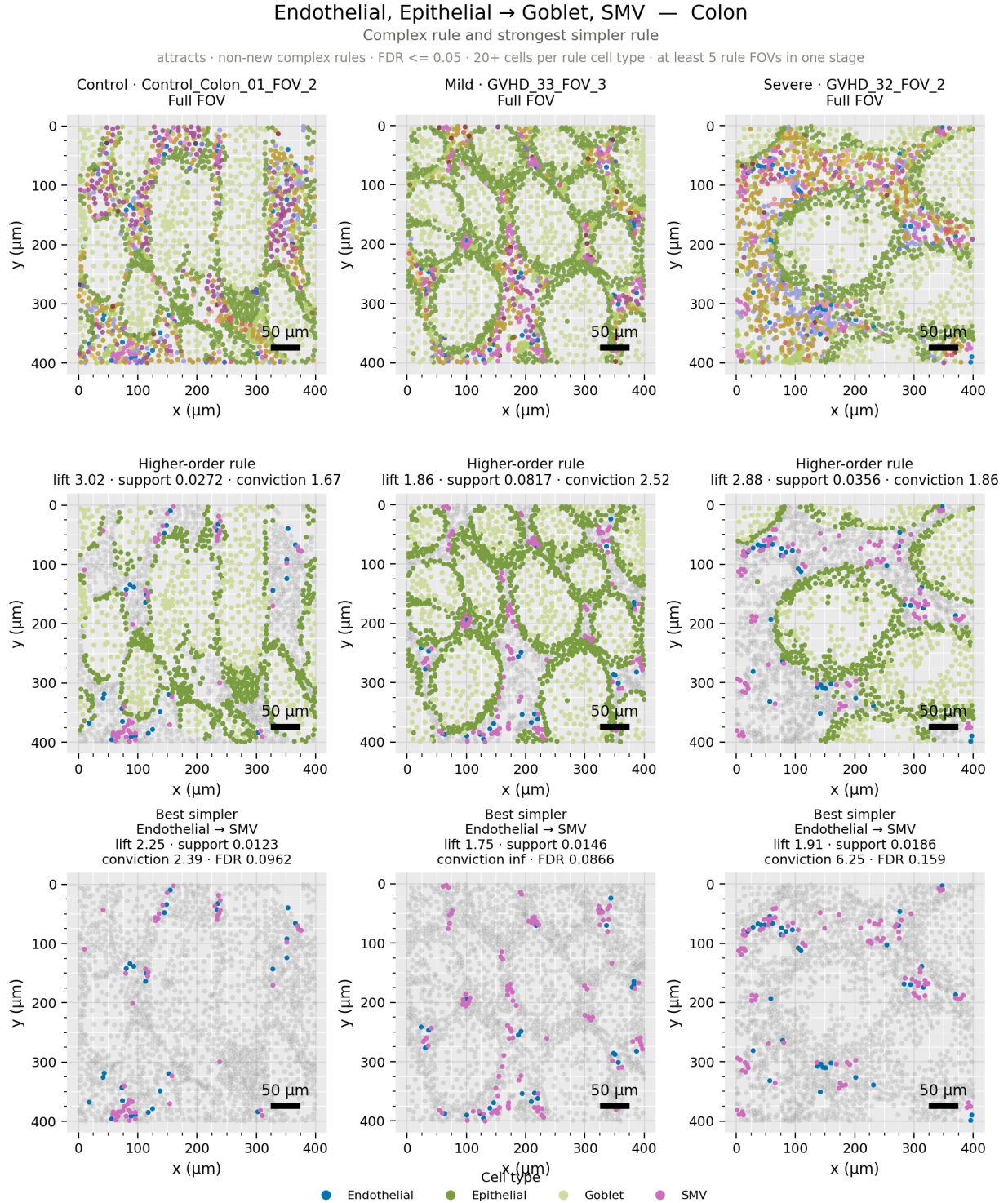


Figure 17: Automatically selected examples show how the four-cell organization occupies tissue beyond the nearly ubiquitous Endothelial → SMV parent.

A separate parentless rule, Epithelial, SMV → Endothelial, supports the same remodeling theme without depending on the non-new Colon result. Figure 18 shows every new occurrence rather than reducing the result to a sentence: 13 FOVs carry the rule, with median Lift 2.76 and a gain floor of 2.30×

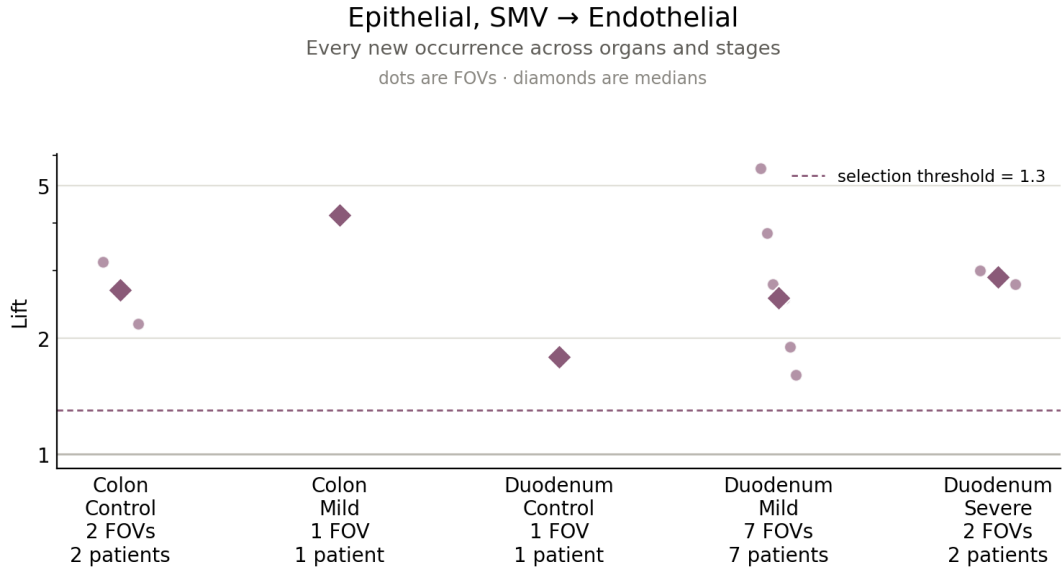


Figure 18: Every FOV in which Epithelial, SMV → Endothelial is classified as new, grouped by organ and stage. Diamonds are medians and the dashed line is the predeclared Lift threshold.

The recurrence plot establishes that this is not a single chosen field; the next panel makes one automatically selected occurrence spatially inspectable.

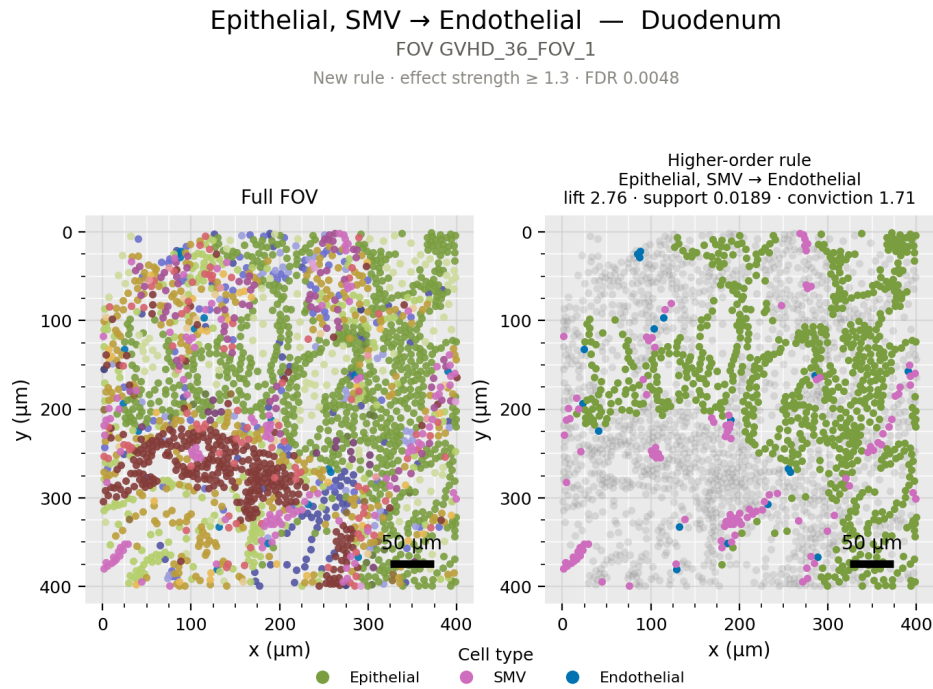


Figure 19: A representative new Epithelial, SMV → Endothelial occurrence. No constituent pair was mined in this FOV.

2.3 A higher-order extension of Duodenal CD8 heterogeneity

The atlas reports “marked inpatient heterogeneity in the prevalence of CD8 T cells and neutrophils in the vicinity of individual crypts and villi.” The pair-rule analysis found increasing Duodenal CD8T–Goblet avoidance. Complex mining adds a macrophage-conditioned version: Goblet → CD8T, Macrophage. The overview first places it in the complete eligible avoidance screen; the availability and strength figures then display the exact denominators, pair coverage and Lift trajectory needed to judge the claim.

Evidence strength: Moderate paper alignment; good FOV denominators but only two Severe patients.

Complex-rule prevalence — Duodenum

Top stage-changing rules that pass the FOV threshold

avoids · non-new complex rules · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

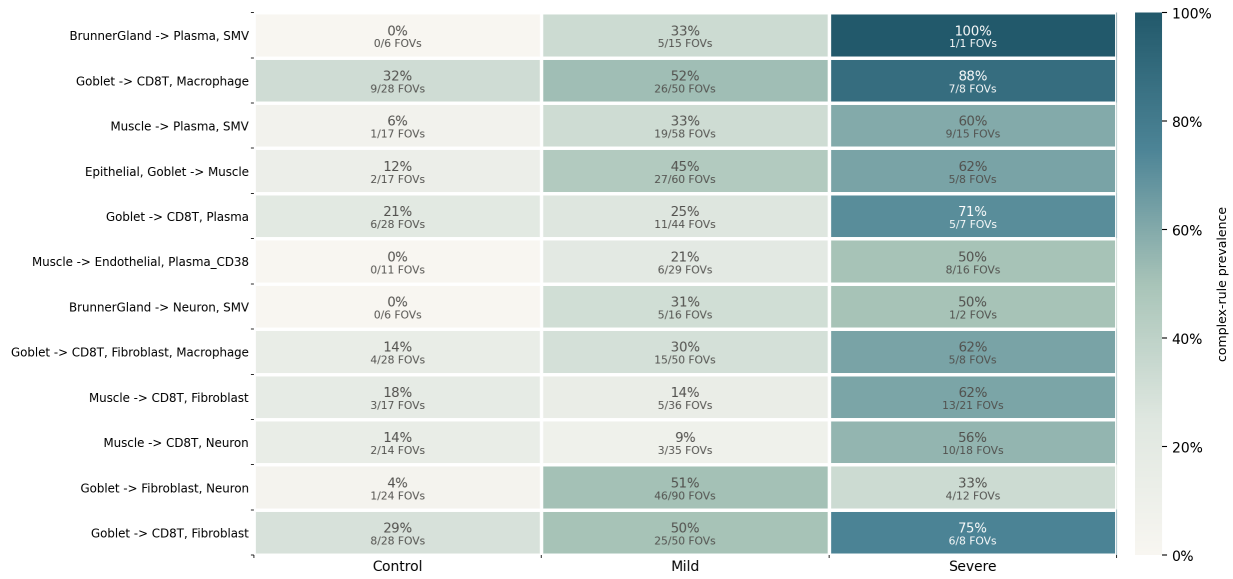


Figure 20: The eligible Duodenal avoidance screen. Multiple CD8T-containing and muscle-containing higher-order rules become more prevalent in Severe disease. Rows with one or two eligible Severe FOVs are visible and are not used as primary claims.

Goblet → CD8T, Macrophage — Duodenum

Cell availability versus higher-order organization

avoids · non-new complex rules · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

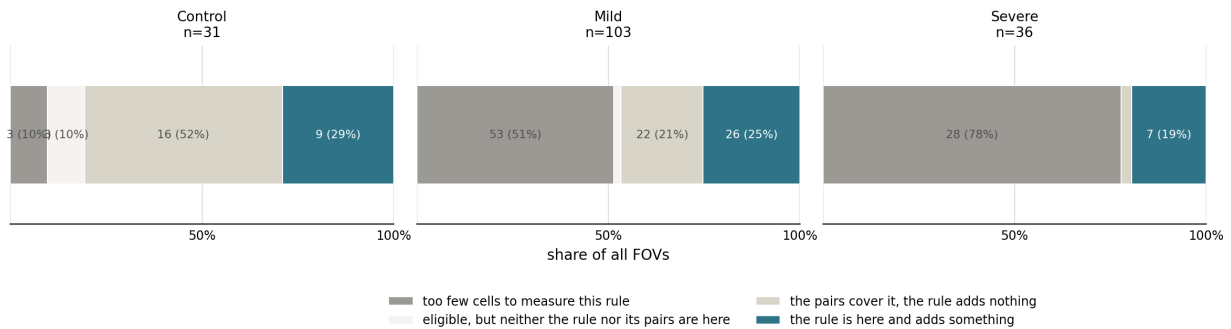


Figure 21: Every Duodenal FOV is assigned to one availability/organization state for Goblet → CD8T, Macrophage. This shows the changing supply of measurable FOVs separately from the fraction carrying the informative higher-order avoidance.

Goblet → CD8T, Macrophage — Duodenum

Prevalence and Lift across disease stages

avoids · non-new complex rules · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

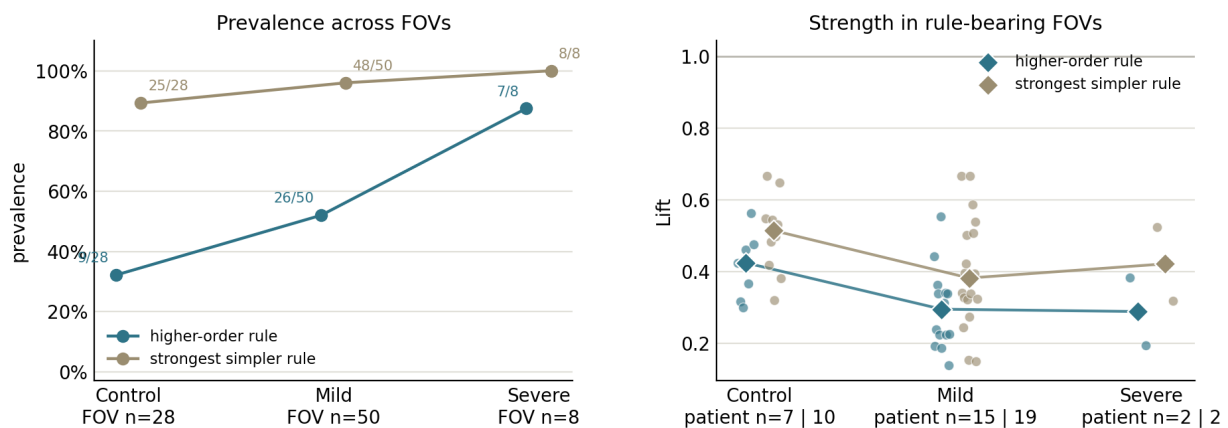


Figure 22: Prevalence and Lift of Goblet → CD8T, Macrophage against its strongest simpler rule. For avoidance, lower Lift means stronger separation.

The figures show complex prevalence rising from 9/28 eligible Control FOVs to 26/50 Mild and 7/8 Severe FOVs, while simpler-rule coverage is already 89–100%. Median complex Lift falls from about 0.42 to 0.29, so the avoidance strengthens numerically. Only two Severe patients contribute to that Lift estimate, so it remains descriptive. The following maps show the actual joint spatial pattern used for the summary.

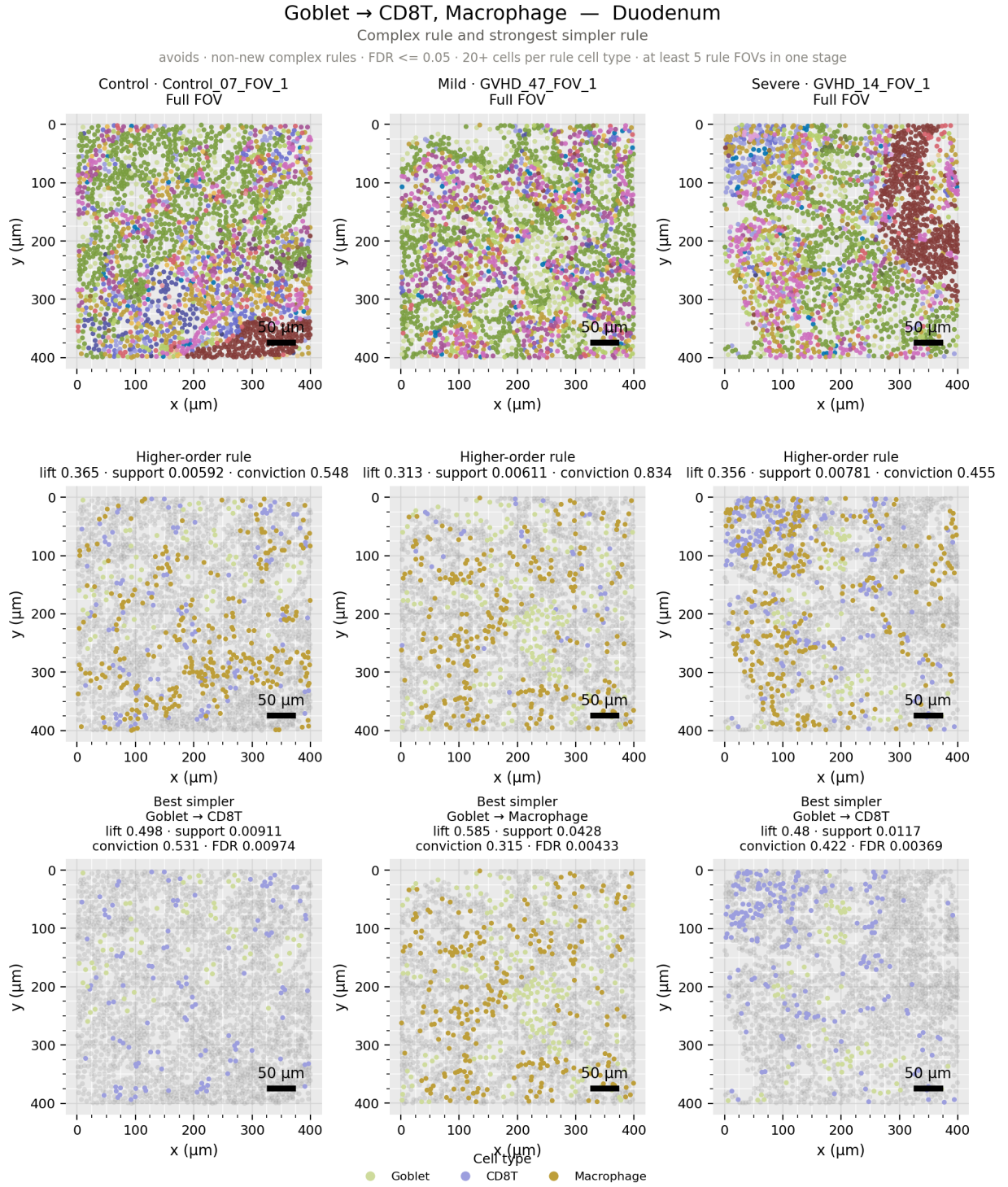


Figure 23: Control, Mild and Severe examples of the macrophage-conditioned CD8T avoidance. These maps make the joint context inspectable but do not identify the atlas's patient-level CD8 subgroup.

3 New biologically interpretable findings

The complete Duodenal new-attraction screen organizes the selected hypotheses into three recurring themes: an epithelial–neural Paneth context, several epithelial–plasma contexts, and smooth-muscle/mesenchymal contexts ending in Endothelial. The overview is shown before the individual rules so that the examples can be judged against the full filtered result rather than as isolated selections.

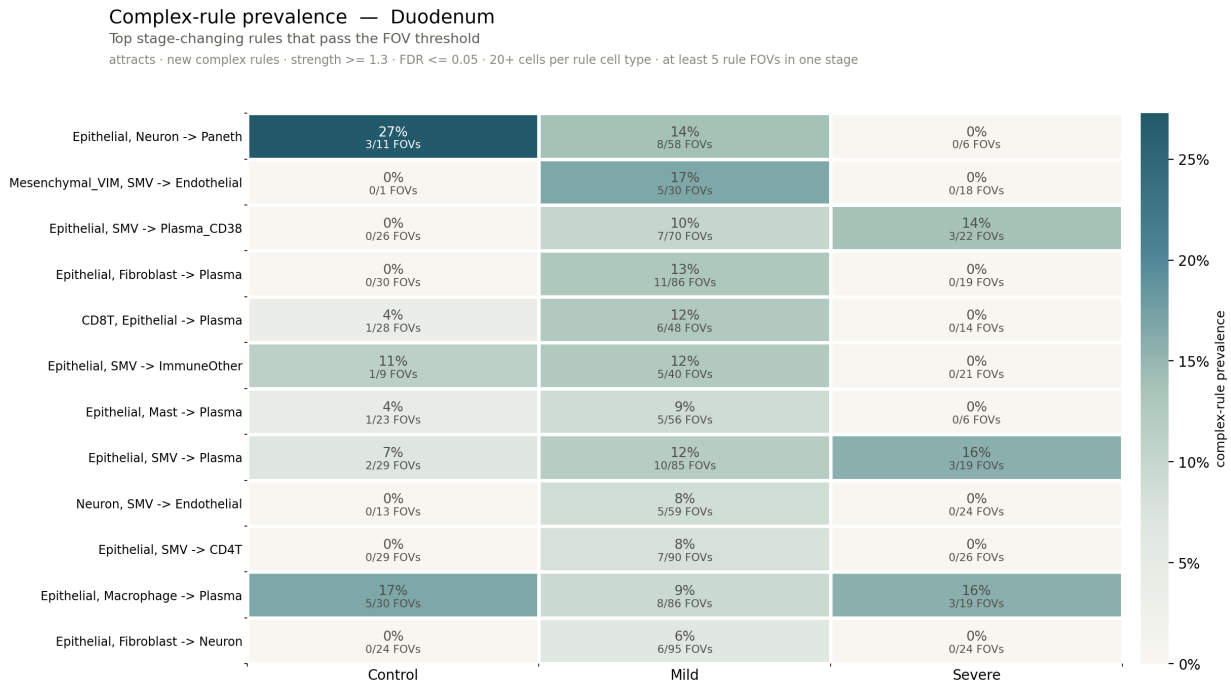


Figure 24: All recurring new Duodenal attractions that pass rule-level FDR, Lift, cell-eligibility and recurrence filters. Exact eligible denominators expose which apparent stage differences can and cannot be trusted.

3.1 A surviving epithelial–neural Paneth niche

Epithelial, Neuron → Paneth is new in 12 FOVs, with median Lift 2.24 and a gain floor of 1.86 $\times$. This extends the atlas observation of Paneth-cell loss: among FOVs that retain enough Paneth cells to measure, Paneth proximity is associated with a joint epithelial–neural context that no simpler mined rule captures. Neuronal signaling can regulate crypt growth, and receptors for that signaling occur in the crypt stem-cell compartment (Muise et al., 2017), making the niche plausible without establishing direct Neuron–Paneth signaling.

Evidence strength: Moderate for the recurring higher-order niche; exploratory mechanism.

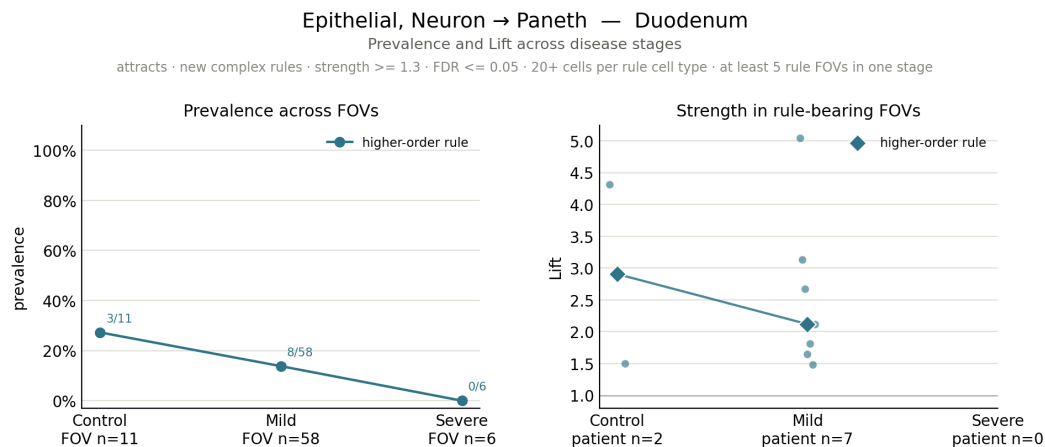


Figure 25: The rule occurs in 3/11 eligible Control and 8/58 Mild FOVs, representing two and seven patients. No Severe rule-bearing FOV passes the filters, so the line is not a validated severity trend.

The prevalence panel establishes recurrence but not morphology. The next figure shows the automatically selected Control and Mild fields, with all non-rule cell types greyed in the highlighted row.

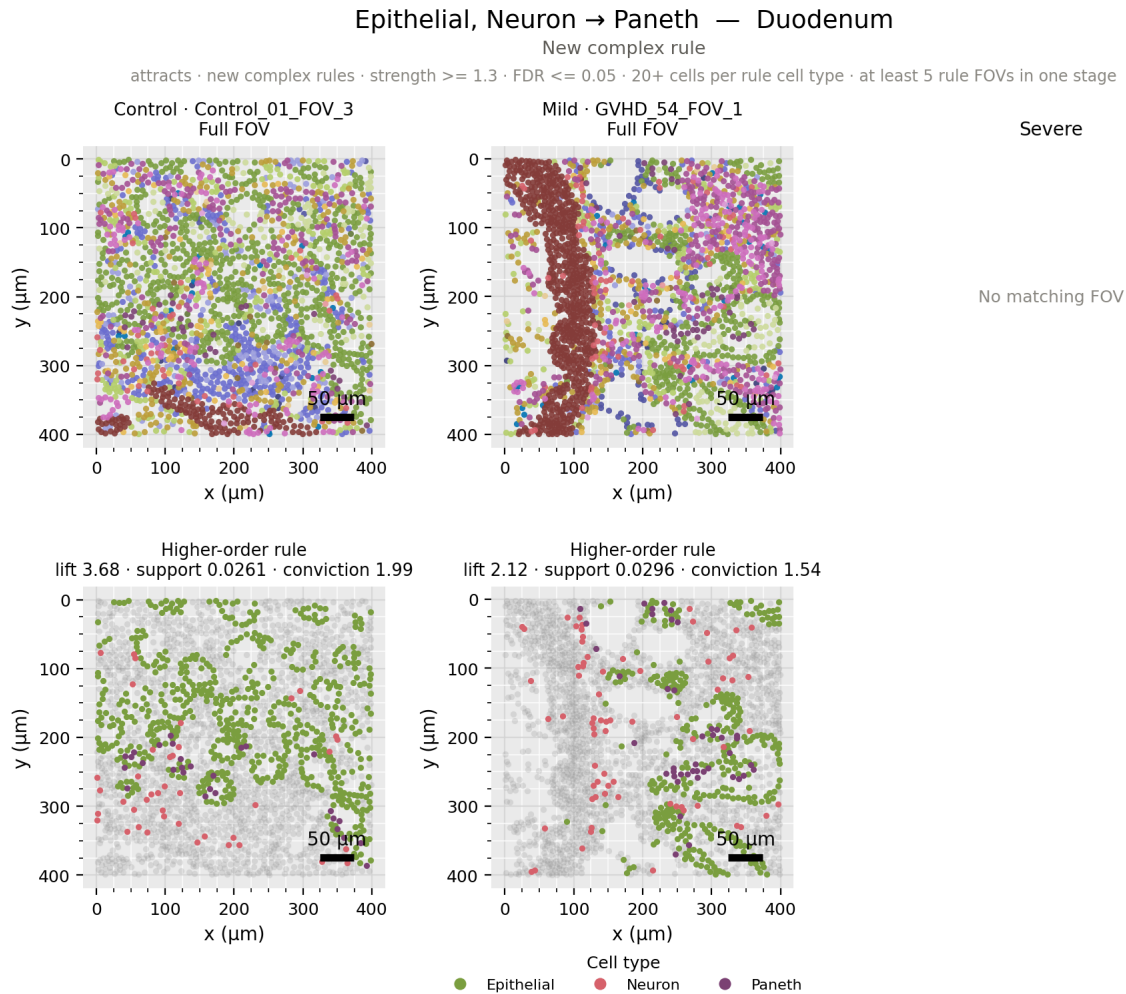


Figure 26: Representative Control and Mild FOVs for the new Paneth rule. The empty Severe column makes the absence of a qualifying example explicit.

3.2 A recurrent neuro–stromal context in Colon

Fibroblast, Muscle → Neuron is a new rule in eight Colon FOVs. It reaches 5/15 eligible Mild FOVs and is also observed in Control and Severe; median Lift is 2.23, 2.67 and 1.50 respectively. The result is biologically coherent with the close organization of enteric nerves and the muscular–stromal compartment, but the patient counts are one, four and two. The robust conclusion is recurrence of a three-cell context, not a stage trend.

Evidence strength: Moderate recurrence; exploratory stage interpretation.

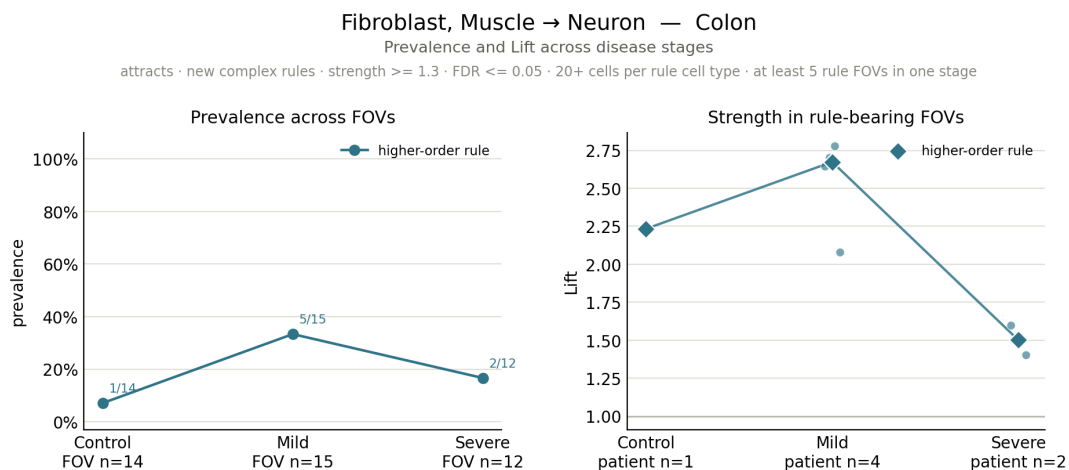


Figure 27: Prevalence and Lift for the new Colon neuro–stromal rule. Exact FOV and patient counts prevent the Mild peak from being overinterpreted.

To verify that the numerical recurrence corresponds to coherent tissue structure, the next panel displays the selected rule-bearing FOV from every stage in which one is available.

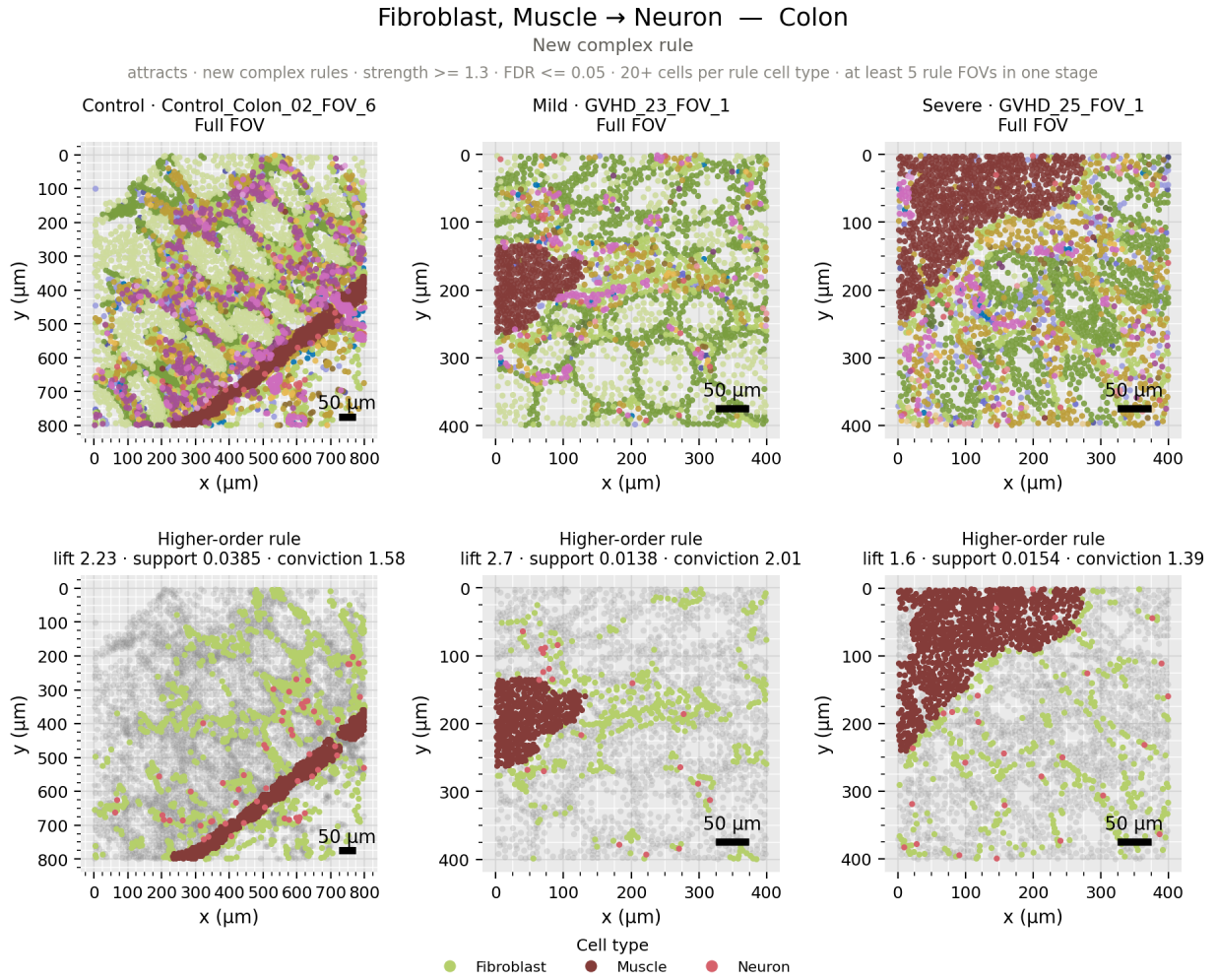


Figure 28: Representative rule-bearing FOVs from all three stages. The highlighted maps reveal a spatially coherent muscular boundary in the selected Control and Mild examples.

3.3 Disease-associated niches among surviving plasma and endothelial cells

Epithelial, SMV → Plasma_CD38 is absent from 26 eligible Control FOVs but present in 7/70 Mild and 3/22 Severe FOVs, with Lift above 1.3. This does not contradict the atlas's overall plasma-cell loss: it suggests that when CD38-positive plasma cells remain, some occupy a disease-associated epithelial–SMV context. Five Mild and two Severe patients carry the rule, so it is a follow-up hypothesis rather than a validated disease trajectory.

Evidence strength: Moderate recurrence with sound eligible denominators; limited patient replication.

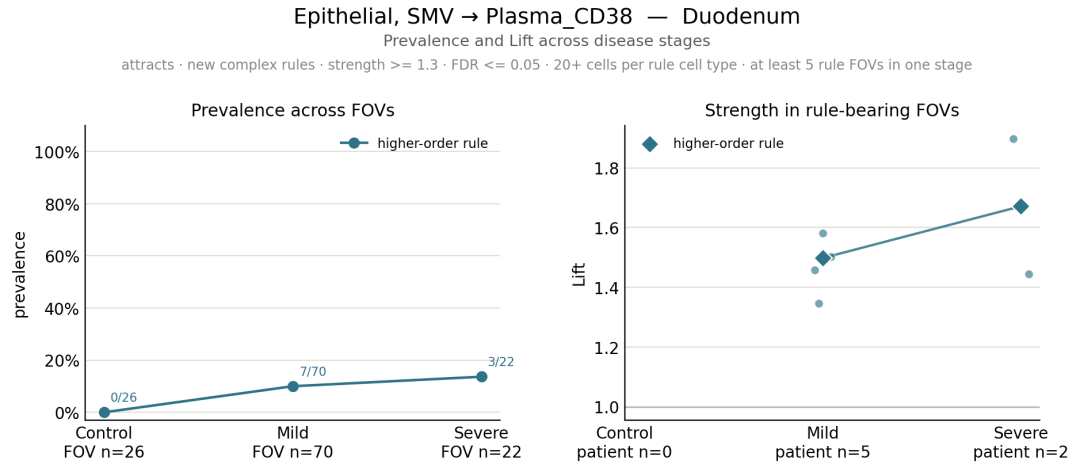


Figure 29: The new Plasma_CD38 context is not observed in eligible Control FOVs and has median Lift 1.50 in Mild and 1.67 in Severe rule-bearing patients.

The next maps show the Mild and Severe occurrences used to ground that stage-level summary; the missing Control example is part of the result, not a plotting omission.

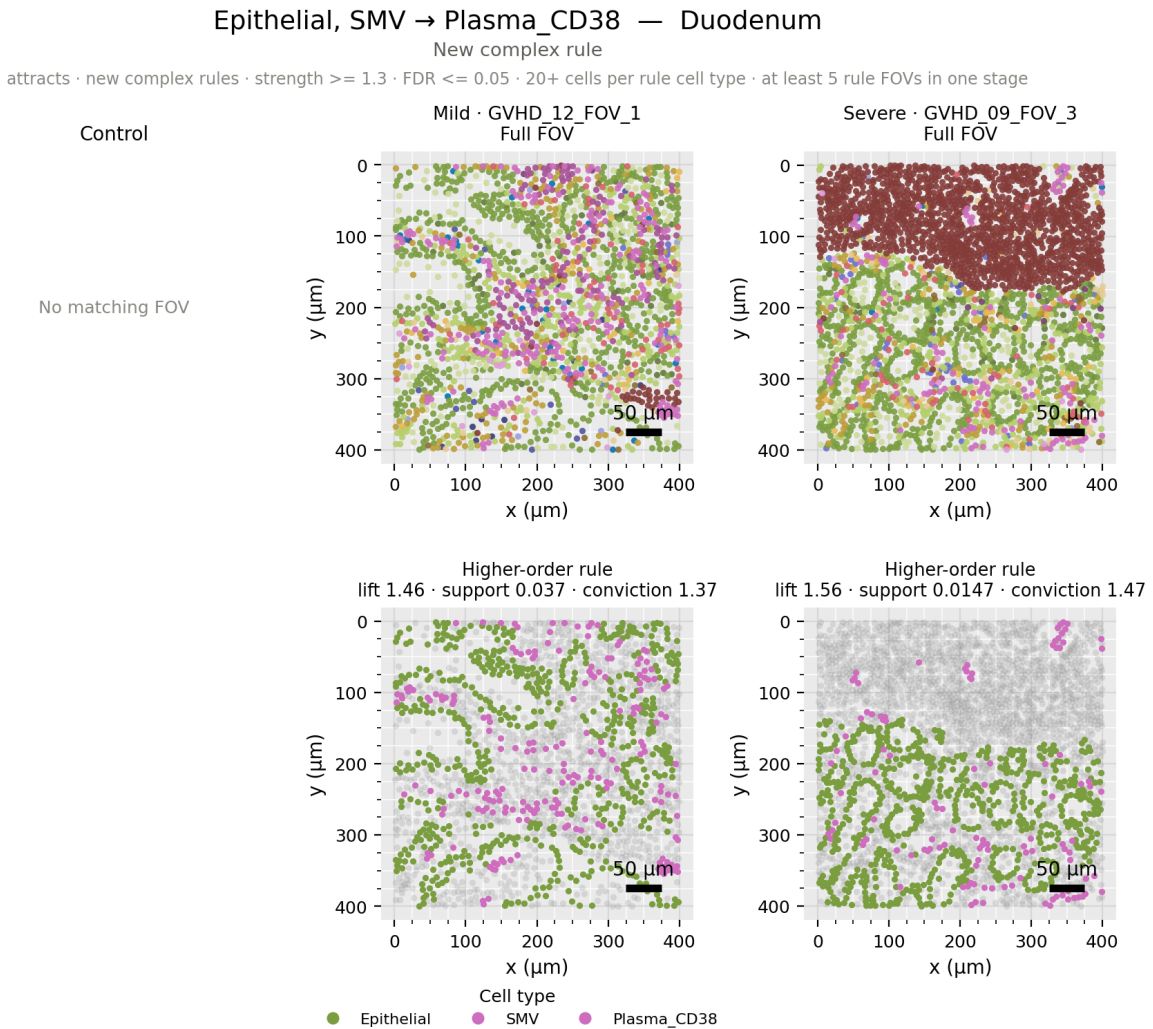


Figure 30: Representative Mild and Severe occurrences. No Control FOV carries the rule despite 26 eligible FOVs.

Mesenchymal_VIM, SMV → Endothelial is detected in five Mild Duodenal FOVs, each from a different patient, with median Lift 2.31. The atlas's fibrosis and stromal disruption make the relationship biologically plausible, but only one Control FOV is eligible. It is therefore evidence for a recurrent Mild context, not evidence for a Control-to-Mild increase.

Evidence strength: Exploratory; five-patient recurrence but an inadequate Control denominator.

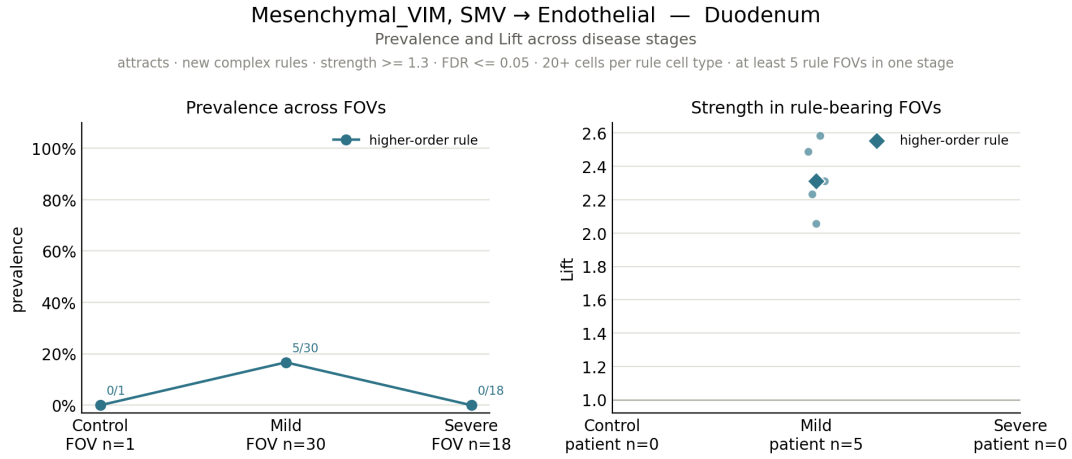


Figure 31: The Mild-only pattern and its eligibility limitation. Control prevalence is 0/1, so the apparent rise must not be interpreted as a stage comparison.

Because the Control denominator prevents a stage claim, the following panel is presented only as spatial confirmation of the recurrent Mild configuration.

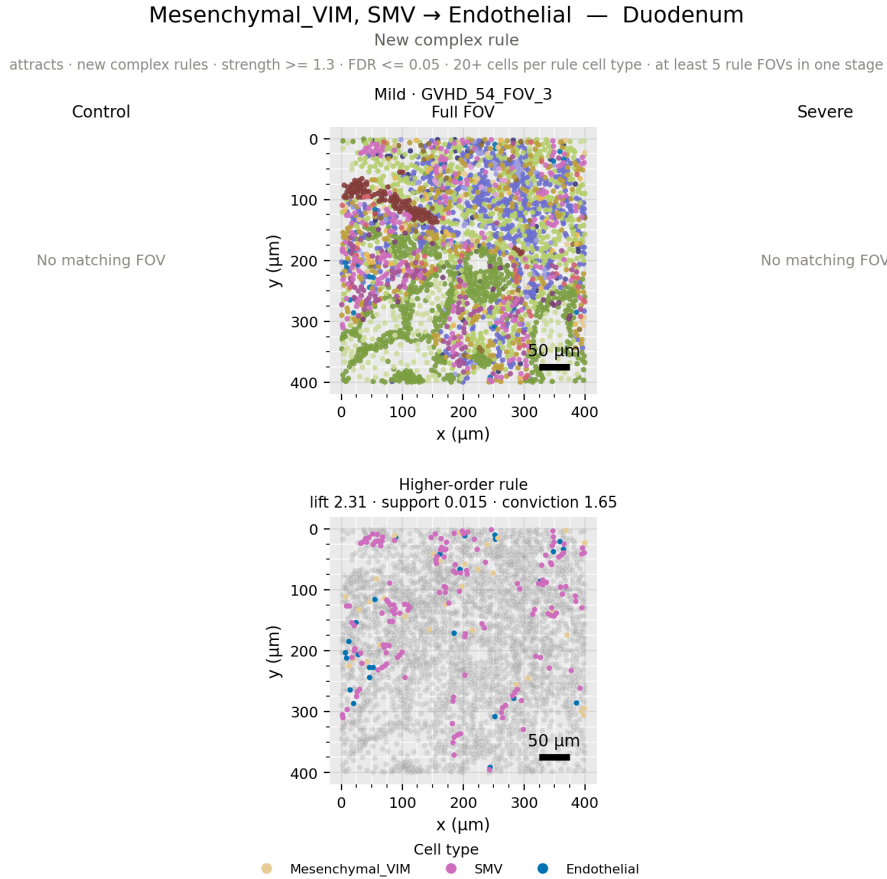


Figure 32: A representative Mild FOV containing the new mesenchymal-smooth-muscle-endothelial context.

3.4 Negative rules: higher-order spatial avoidance

Avoidance is analyzed as a complete result family, using the same FDR, cell-eligibility, strength and recurrence filters as attraction. No new Colon avoidance reaches five FOVs in one eligible stage. The retained Duodenal screen is small and coherent: all five rules contain Muscle or SMV, but no stage prevalence exceeds 11%. This supports a recurring muscle-centered family while arguing against a strong new-avoidance stage claim.

Complex-rule prevalence — Duodenum

Top stage-changing rules that pass the FOV threshold

avoids · new complex rules · strength ≥ 1.3 · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

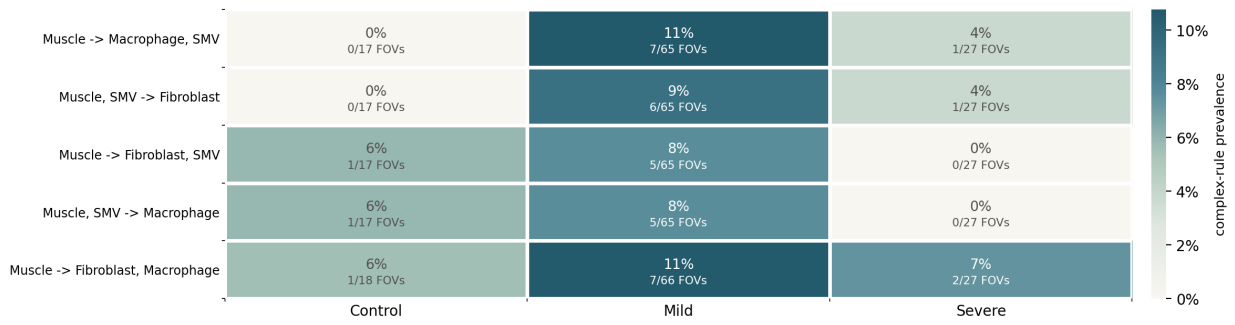


Figure 33: All recurring new Duodenal avoidances. The common muscle/SMV theme is biologically interesting, but the low prevalence keeps the stage interpretation exploratory.

The new avoidance Muscle → Fibroblast, Macrophage recurs in 17 FOVs, with median inverse Lift 1.40. A representative Duodenal FOV has Lift 0.717, support 0.0585, conviction 0.576 and FDR 0.00656. This joins the Fibroblast/Muscle/Neuron finding in pointing to the muscular–stromal compartment, but in the opposite spatial direction. Muscularis macrophages participate in gut neuromuscular homeostasis (Müller et al., 2014); the present rule does not identify that macrophage subtype and should be followed up by cell state or marker analysis.

Evidence strength: Moderate recurrence; biologically plausible but subtype-agnostic.

Muscle → Fibroblast, Macrophage — Duodenum

Prevalence and Lift across disease stages

avoids · new complex rules · strength ≥ 1.3 · FDR ≤ 0.05 · 20+ cells per rule cell type · at least 5 rule FOVs in one stage

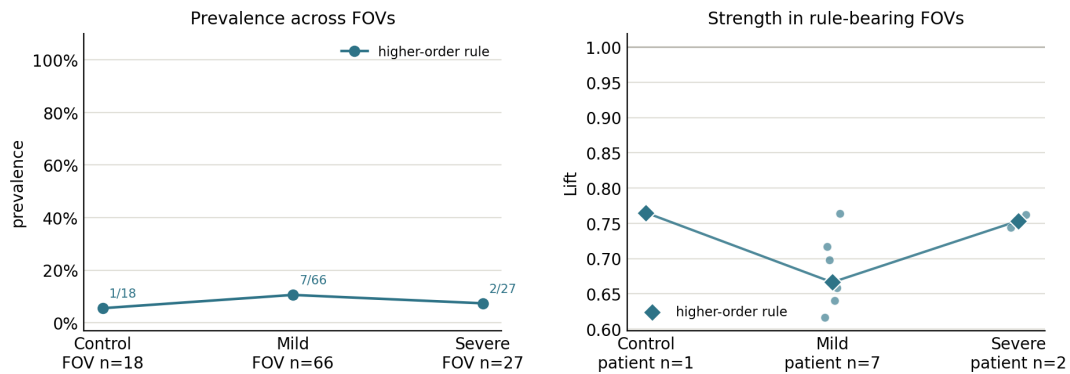


Figure 34: Eligible stage prevalence and Lift for the new Muscle → Fibroblast, Macrophage avoidance. The exact FOV and patient counts show both its recurrence and why its stage trend remains exploratory.

The recurrence statistics motivate inspection of stage-specific fields. Since this is a new rule, the spatial panel correctly contains no simpler-rule row.

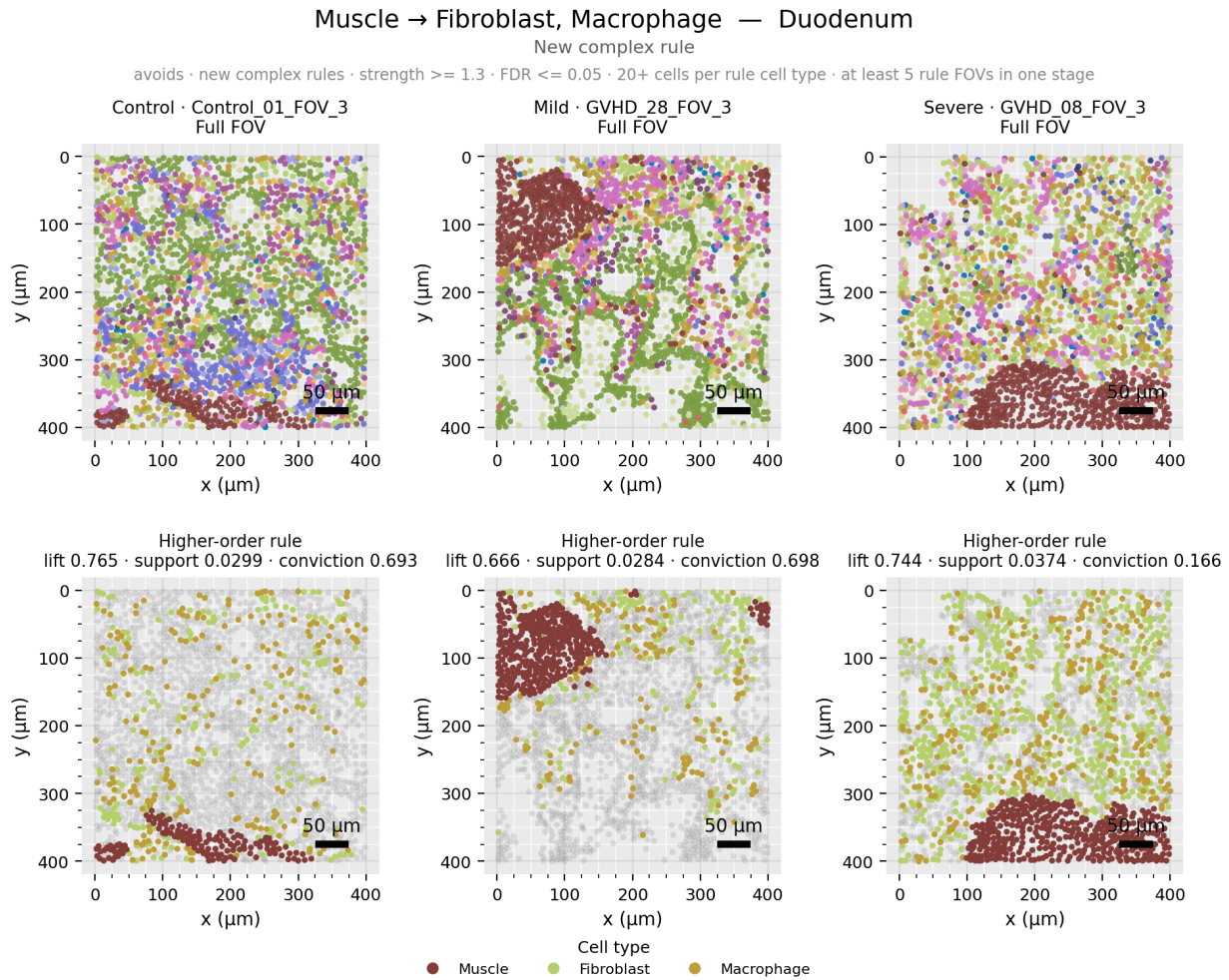


Figure 35: Automatically selected stage examples of the new Muscle → Fibroblast, Macrophage avoidance. Each available column shows the full field and the higher-order rule.

A complementary non-new avoidance, Muscle → Plasma, SMV, becomes substantially more prevalent in disease while its simpler-rule coverage remains high. Its complex Lift rises from 0.04 in Control to 0.17 in Mild and 0.11 in Severe: the avoidance is much more frequent in disease but is not stronger than the single Control estimate. The next figure shows both summaries and both rules, preventing prevalence from being confused with effect strength. Only one Control patient carries the complex rule, so the Lift trajectory is descriptive.

Evidence strength: Strong FOV-level prevalence contrast; Control strength is based on one patient.

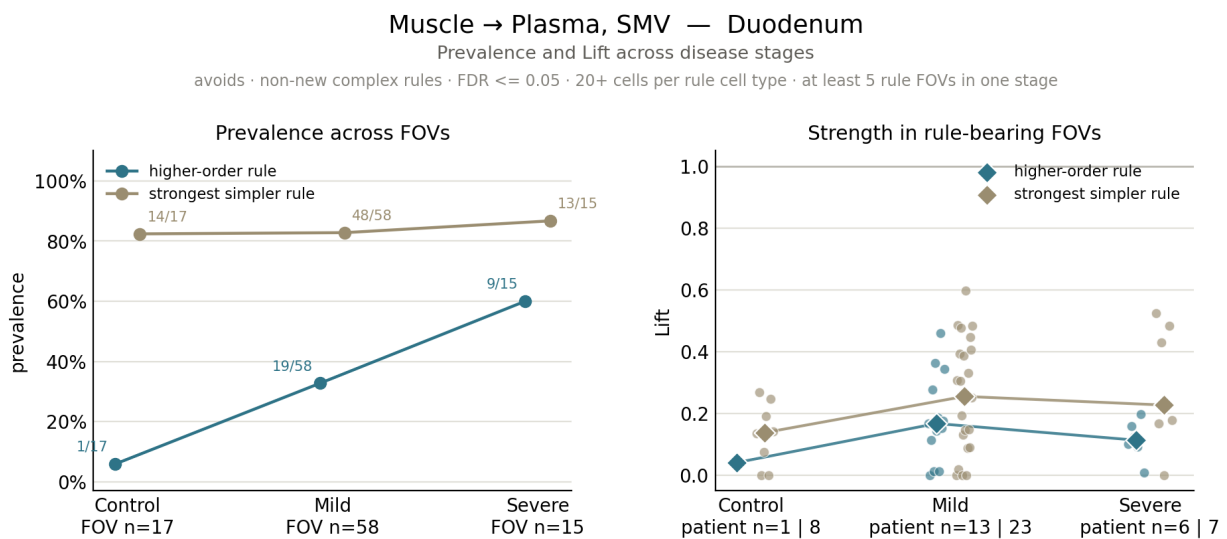


Figure 36: The non-new Duodenal Muscle → Plasma, SMV avoidance rises from 1/17 eligible Control FOVs to 19/58 Mild and 9/15 Severe FOVs; its strongest simpler rule is shown in both panels.

A distinct non-new rule, Muscle $\rightarrow$ CD8T, Fibroblast, has adequate eligible denominators in every stage. Complex-rule prevalence rises from 3/17 Control and 5/36 Mild FOVs to 13/21 Severe FOVs, whereas the strongest simpler rule is more common throughout. At the same time, complex Lift rises from 0.20 to 0.45, meaning the avoidance becomes weaker among rule-bearing FOVs. This divergence is important: prevalence and effect strength are different biological summaries and neither should substitute for the other.

Evidence strength: Strong descriptive example of prevalence–strength divergence; six Severe patients.

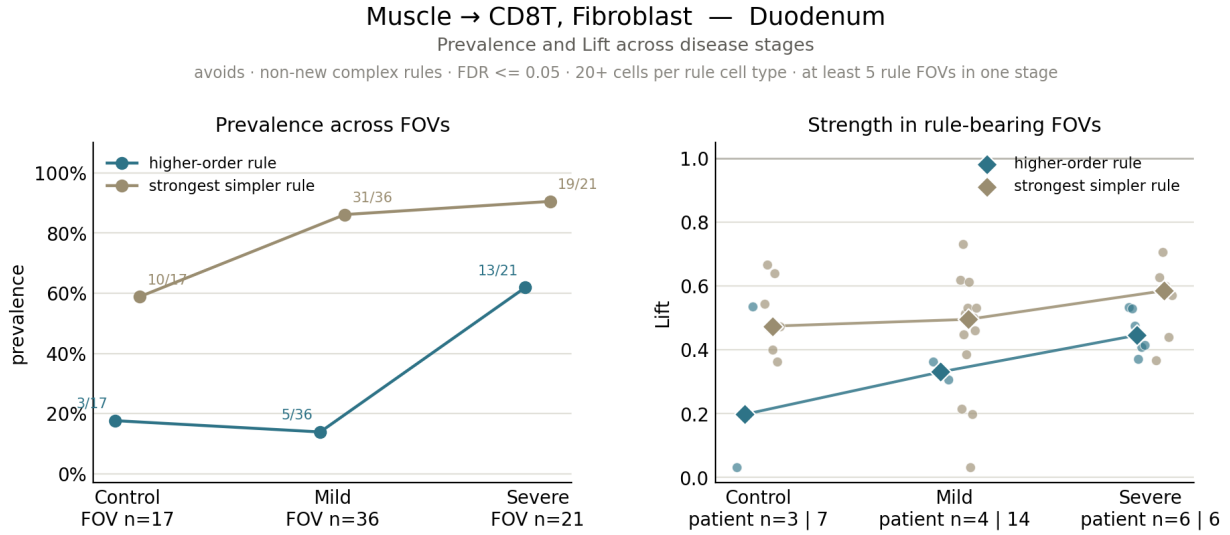


Figure 37: The complex rule becomes more prevalent but less strongly avoidant across stages. The side-by-side panels prevent the prevalence increase from being misread as stronger spatial separation.

The final maps show the stage examples behind both lines and retain the strongest simpler rule so the reader can inspect what the extra Fibroblast context adds.

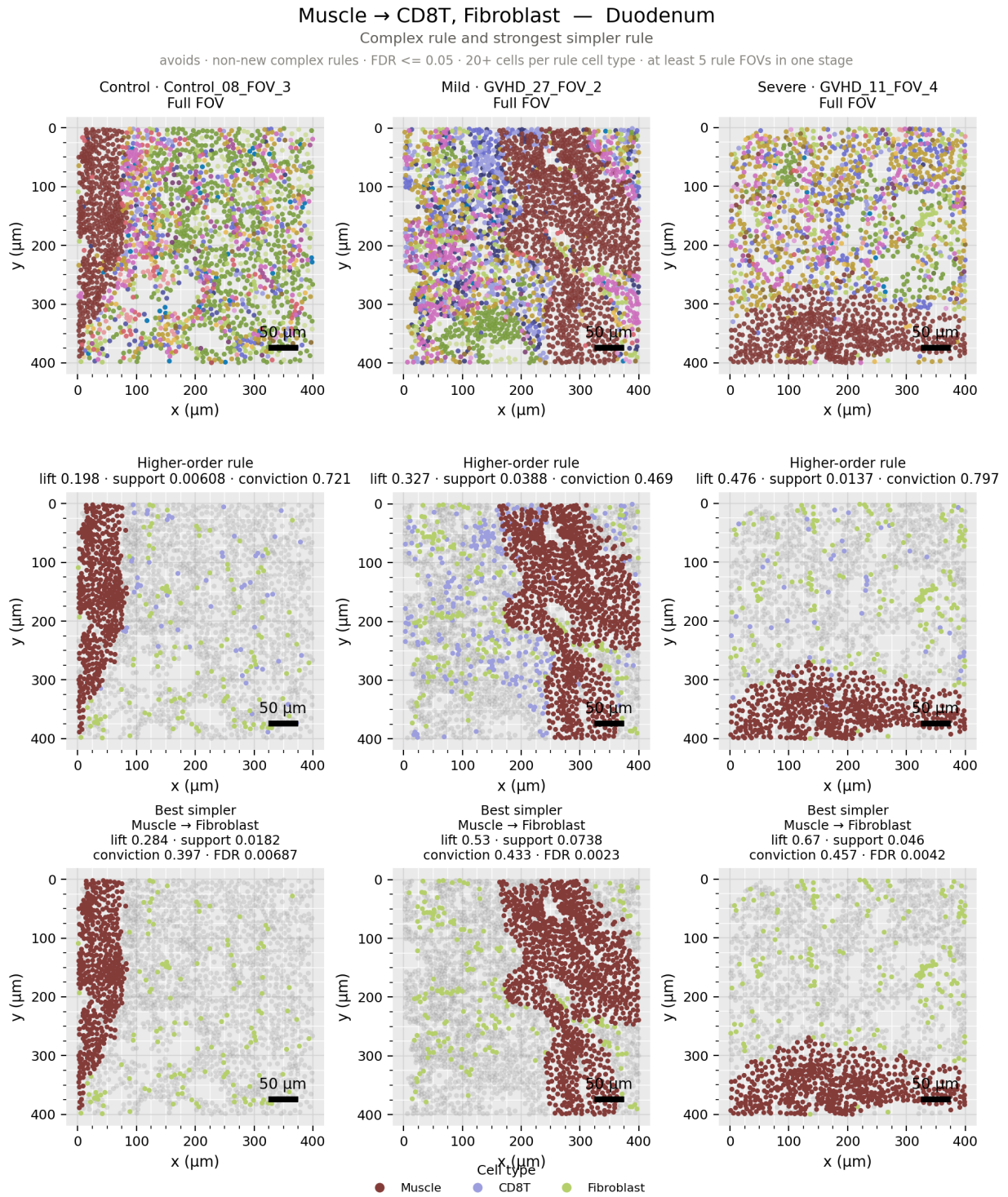


Figure 38: Representative Control, Mild and Severe examples of the complex and strongest simpler rule.